MATHEMATICS PROJECT WORK RECORD :: FA-1

Topic: Sets and Venn Diagrams

1. PRELIMINARY INFORMATION

Name of the Student:	.....
Class & Section:	.....
Roll Number:	.....
Subject:	**Mathematics**
Topic:	**Sets and Venn Diagrams**

2. AIM

To survey 30 classmates on their sports preferences (Cricket and Badminton), represent the collected data using set operations, and verify set relationships geometrically using a Venn diagram.

3. OBJECTIVES

- Collect primary data regarding two sports choices from 30 students.
- Represent the responses using standard set notation (Union, Intersection, and Difference).
- Verify the fundamental formula: $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.
- Illustrate regions and subsets clearly using a Venn diagram.

4. MATERIALS REQUIRED

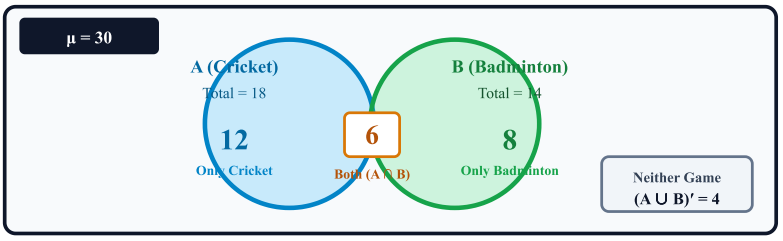
Project paper, compass, ruler, pencil, pen, and Class 10 Mathematics textbook.

5. PROCEDURE

1. Surveyed 30 students (Universal Set μ, roll numbers 1 to 30) regarding their preferences.
2. Defined Set A as students playing Cricket and Set B as students playing Badminton.
3. Determined counts of students playing only Cricket, only Badminton, both, or neither.
4. Recorded the cardinal values: $n(\mu)$, $n(A)$, $n(B)$, $n(A \cap B)$, and $n((A \cup B)')$.
5. Constructed a formal two-circle Venn diagram inside a rectangular universal set box.

6. OBSERVATIONS AND TABULATION

Category	Notation	Student Roll Numbers	Cardinal Number
Total Surveyed (Universal Set)	μ	{1, 2, 3, ..., 30}	n(μ) = 30
Students playing Cricket	A	{1, 2, 4, 5, 7, 8, 10, 11, 13, 14, 16, 17, 19, 20, 22, 23, 25, 26}	n(A) = 18
Students playing Badminton	B	{4, 5, 7, 8, 10, 11, 12, 15, 18, 21, 24, 27, 28, 29}	n(B) = 14
Playing both games	A ∩ B	{4, 5, 7, 8, 10, 11}	n(A ∩ B) = 6
Playing at least one game	A ∪ B	—	n(A ∪ B) = 26
Playing neither game	(A ∪ B)′	{3, 6, 9, 30}	n((A ∪ B)′) = 4



Verification: n(A ∪ B) = n(A) + n(B) − n(A ∩ B) = 18 + 14 − 6 = **26**.

7. CONCLUSION

- Venn diagrams provide a clear visual tool to represent grouped data and intersections.
- The cardinal formula n(A ∪ B) = n(A) + n(B) − n(A ∩ B) was verified accurately with survey data.

8. ACKNOWLEDGEMENT

I express my sincere thanks to my **Mathematics Teacher** for continuous guidance and my classmates for their cooperation during data collection.

9. RESOURCES

Telangana SCERT Class 10 Mathematics Textbook and classroom survey records.

Signature of Student
Date: ____/____/2026

Signature of Mathematics Teacher
Verified